Carbohydrate Ingestion during Prolonged Cycling Improves Next-Day Time Trial Performance and Alters Amino Acid Concentrations

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ABSTRACT

CLAUSS, M., Ø. SKATTEBO, M. RASEN DÆHLI, T. DITTA VALSDOTTIR, N. EZZATKHAH BASTANI, E. IVAR JOHANSEN, K. JENSEN KOLNES, B. STEEN SKÅLHEGG, and J. JENSEN. Carbohydrate Ingestion during Prolonged Cycling Improves Next-Day Time Trial Performance and Alters Amino Acid Concentrations. *Med. Sci. Sports Exerc.*, Vol. 55, No. 12, pp. 2228–2240, 2023. **Introduction:** Exercise with low carbohydrate availability increases protein degradation, which may reduce subsequent performance considerably. The present study aimed to investigate the effect of carbohydrate ingestion during standardized exercise with and without exhaustion on protein degradation and next-day performance. **Methods:** Seven trained male cyclists ($\dot{V}O_{2max}$ 66.8 ± 1.9 mL·kg^{-1}·min^{-1}; mean ± SEM) cycled to exhaustion (~2.5 h) at a power output eliciting 68% of $\dot{V}O_{2max}$ ($W_{68\%}$). This was followed by repeating 1-min work/1-min recovery intervals at 90% of $\dot{V}O_{2max}$ ($W_{90\%}$) until exhaustion. During $W_{68\%}$, cyclists consumed a placebo water drink (PLA) the first time and a carbohydrate drink (CHO), 1 g carbohydrate·kg^{-1}·h^{-1}, the second time. The participants performed the same amount of work under the two conditions, separated by at least 1 wk. A standardized diet was provided to the participants so that the two conditions were isoenergetic. To test the impact of carbohydrates on recovery, participants completed a time trial (TT) the next day. **Results:** Carbohydrate ingestion maintained carbohydrate availability during $W_{68\%}$ and $W_{90\%}$: total carbohydrate oxidation was significantly higher in CHO ($P = 0.022$), and plasma glucose concentration was maintained compared with PLA ($P = 0.025$). Next-day performance during TT was better after CHO ingestion (CHO, 41:49 ± 1:38 min; PLA, 42:50 ± 1:46 min; $P = 0.020$; effect size $d = 0.23$, small), as was gross efficiency (CHO, 18.6% ± 0.3%; PLA, 17.9% ± 0.3%; $P = 0.019$). Urinary nitrogen excretion ($P = 0.897$) and urinary 3-methylhistidine excretion ($P = 0.673$) did not significantly differ during the study period. Finally, tyrosine and phenylalanine plasma concentrations increased in PLA but not in CHO ($P = 0.018$). **Conclusions:** Carbohydrate ingestion during exhaustive exercise reduced deterioration in next-day performance through reduced metabolic stress and development of fatigue. In addition, some parameters point toward less protein degradation, which would preserve muscle function. **Key Words:** PROTEIN DEGRADATION, NITROGEN BALANCE, 3-METHYLHISTIDINE, BRANCHED-CHAIN AMINO ACIDS, GLUCOSE

Carbohydrates and fat are the primary energy substrates during endurance exercise. However, their relative contributions depend on the intensity and duration of exercise (1,2). During moderate- and high-intensity exercise, carbohydrates provide the majority of the energy supply, but carbohydrate reserves are limited (3).

Protein usually contributes only marginally to energy production (4–7). However, when exercising with low carbohydrate availability, protein degradation is more than double when exercising with high carbohydrate availability (6,8). Cycling to exhaustion is known to deplete carbohydrate reserves, with participants exhibiting low muscle glycogen levels (3) and low plasma glucose concentrations at exhaustion (3,9,10). This form of exercise also induces high nitrogen excretion (3,9,10), which is indicative of protein degradation, most probably from muscle protein (11). When proteins are degraded, the amino groups are mainly converted to urea and excreted in the urine. Next-day performance has been found to increase when nitrogen excretion is lower (9,10).

Interestingly, ingesting carbohydrates during exercise is an established way to maintain carbohydrate availability (12–14).

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This has been shown when plasma glucose concentration dropped significantly when participants cycled to exhaustion while drinking water compared with a glucose drink (12). Carbohydrate ingestion also delayed exhaustion by 60 min. However, to what extent carbohydrate ingestion influences protein degradation during strenuous endurance cycling has not been clarified.

The aim of the present study was to investigate the effect of carbohydrate ingestion during standardized exercise with and without exhaustion on protein degradation and next-day performance. Therefore, participants cycled to exhaustion without an energy supply and repeated the same amount of work a week later while consuming a carbohydrate drink to avoid voluntary exhaustion. We hypothesized that ingesting carbohydrates during standardized exercise with and without exhaustion would maintain carbohydrate availability, reduce protein degradation, and reduce next-day performance degradation.

MATERIALS AND METHODS

Participants

Seven male trained endurance cyclists (mean ± SEM; age, 29.3 ± 2.6 yr; height, 1.83 ± 0.03 m; body mass, 78.0 ± 3.8 kg; $\dot{V}O_{2max}$, 5.2 ± 0.2 $L \cdot min^{-1}$, 66.8 ± 1.9 $mL \cdot kg^{-1} \cdot min^{-1}$) were recruited for the study. To physically take part in this study, the participants had to train in cycling more than three times a week for the last 6 months. They achieved a weekly training volume of 10.0 ± 0.9 h. Participants were informed about the study before providing their written informed consent. The Norwegian School of Sport Sciences Ethics Committee (application 121-241019) and the Norwegian Centre for Research Data (reference number 253131) approved the study, which conformed to the standards set by the latest revision of the Declaration of Helsinki.

Study design. The study was completed in a single-blinded, balanced, crossover experimental design, as depicted in Figure 1. The participants performed two initial tests to determine physiological parameters and for familiarization. The first time, they performed the 2-d exercise protocol (day 1 and day 2) while consuming a water placebo. They then repeated the exercise protocol at least 1 wk later, while consuming a carbohydrate drink.

Physiological testing. On the first visit to the laboratory, participants performed an incremental test on a cycle ergometer (Lode Excalibur; Lode, Groningen, The Netherlands) during which the relationship between work rate (W) and oxygen uptake ($\dot{V}O_2$) was established. This test and all tests in this study were performed in a controlled environment, free from distractions, under similar ambient conditions (18°C–19°C). The saddle and handle positions were individually adjusted and recorded for replication on the remaining test days.

Participants started the incremental test at 200 W, and the load was increased by 25 W every 5 min. $\dot{V}O_2$ was measured online (Oxycon Pro, Jager Instruments, Hoechberg, Germany) over 150 s (from ~2.5–5 min at each intensity level), and capillary blood samples were collected for lactate analysis (Biosen

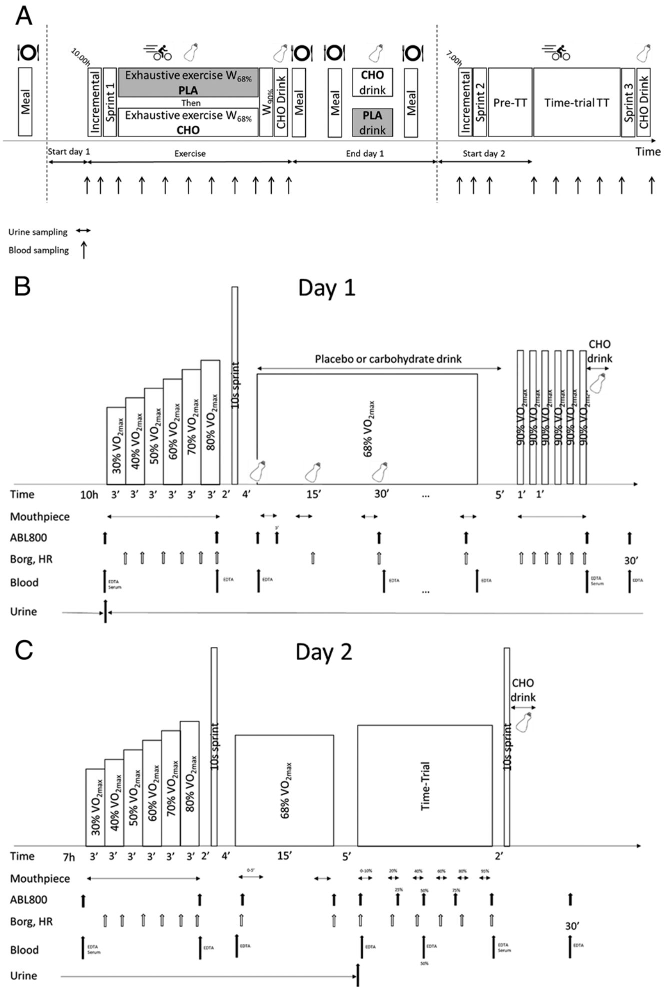

FIGURE 1—Design of the interventions. A, The protocol was completed twice in a single-blinded, balanced, crossover experimental design. On day 1, the participants reported to the laboratory at 9:30 AM and at 10:00 AM prepared for an exhaustive training protocol. After an incremental fat oxidation profile warm-up, participants performed a 10-s sprint (sprint 1), followed by an exhaustive cycling exercise at 68% of $\dot{V}O_{2max}$ ($W_{68\%}$) for a total of 147 ± 17 min. During the exhaustive cycling exercise, the participants drank a flavored water placebo drink (PLA) in week 1 and a similar-flavored carbohydrate drink (CHO) in week 2. After exhaustion, 1-min intervals at 90% of $\dot{V}O_{2max}$, interspersed with a minute of recovery, were performed until exhaustion. After the exercise, the diet was standardized during the 17-h recovery period. On day 2, the participants reported to the laboratory at 6:30 AM. The incremental fat oxidation profile warm-up started at 7:00 AM, followed by a 10-s sprint (sprint 2), a 15-min cycling exercise at $W_{68\%}$ (Pre-TT), a time trial performance test (TT), a second 10-s (sprint 3), and a carbohydrate drink. The names of the urine collection periods are indicated in the figure. B, Description of the cycling protocol on day 1. C, Description of the cycling protocol on day 2.

C-Line Lactate analyzer, EKF Diagnostics, UK). The incremental test was terminated after approximately four to five steps when the blood lactate concentration exceeded 4 mM. Participants selected a pedaling frequency above 75 rpm, and the self-selected pedaling frequency was used throughout the study. After 10 min of rest, maximal oxygen uptake ($\dot{V}O_{2max}$) was measured. Participants started the $\dot{V}O_{2max}$ test at the second to last step of the first test, and intensity was increased by 25 W steps every 60 s until voluntary exhaustion. $\dot{V}O_2$

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was measured continuously, and the mean of the two highest consecutive 30-s epochs was defined as the maximal oxygen uptake ($\dot{V}O_{2max}$).

Linear regression analysis from this incremental cycling test was used with $\dot{V}O_{2max}$ to determine the work rates corresponding to 68% of $\dot{V}O_{2max}$ ($W_{68\%}$) and 90% of $\dot{V}O_{2max}$ ($W_{90\%}$).

Familiarization session. The participants returned to the laboratory 2 to 4 d after the physiological testing to complete a familiarization session. They performed the exercise protocol's second day (day 2), including the performance test. First, participants completed an incremental fat oxidation profile warm-up (6 × 3 min at 30%, 40%, 50%, 60%, 70%, and 80% of $\dot{V}O_{2max}$ in this order). $\dot{V}O_2$ was measured continuously during this warm-up. After 2 min, participants carried out a 10-s maximal sprint. All sprints during the study were all-out efforts using a torque factor of 0.7 $N\cdot m\cdot kg^{-1}$. After 4 min of recovery, participants performed a 15-min exercise at a fixed intensity corresponding to 68% of $\dot{V}O_{2max}$. $\dot{V}O_2$ was measured from 0 to 5 min and 12 to 15 min. Calculations were made after this session to ensure that the work rate to achieve a $\dot{V}O_2$ corresponding to 68% of $\dot{V}O_{2max}$ was correct. If not, the work rate was adjusted for the next time to meet the expected value. After a 5-min recovery, participants performed TT, during which they had to complete a specific amount of mechanical work (10). The total work to be completed was equivalent to 30 min (1800 s) at a work rate corresponding to 100% of $\dot{V}O_{2max}$ [work output (kJ) = power output at $\dot{V}O_{2max}$ (W) × 1800 s]. TT started at $W_{68\%}$, but the participants could adjust the work rate during the TT on a panel mounted on the bike as they preferred. Time was blinded during TT, but accumulated work was shown on a screen in front of the participants. Participants were also informed when they had completed 10%, 20%, 40%, 60%, 80%, 95%, and 100% of TT. Two minutes after the end of TT, participants performed a second 10-s maximal sprint.

Diet intervention before the exercise protocol. Participants recorded their dietary intake 24 h before the first intervention and were asked to follow the same diet before the next intervention. Participants were asked to avoid consuming meat, fish, and soy-based products 3 d before the start of each intervention to control for 3-methylhistine concentration (15). Participants did not consume meat, fish, and soy-based products during the study protocol. Participants received a standardized dinner the day before and were asked to consume it at 2100 h. This meal was composed of a vegetarian pizza (Grandiosa Mandagspizza, Grandiosa, Norway). Additional cheese or fruit cocktail was given to adjust the macronutrient content of the meal to body weight. This meal adjustment of macronutrient and energy content to body weight was performed for every meal. The macronutrient content of this meal was carbohydrate 2.03 $g\cdot kg^{-1}$, protein 0.81 $g\cdot kg^{-1}$, and fat 0.57 $g\cdot kg^{-1}$ (69.7 $kJ\cdot kg^{-1}$). The energy intake and the content of macronutrients for the different standardized meals and supplements during the study protocol are described in Table 1. No exercise or only light exercise was allowed for a minimum of 24 h before the exhaustive exercise.

Day 1. Participants reported to the laboratory at 0930 h after a 12- to 14-h fast. A Venflon catheter was inserted into an antecubital vein. Participants were weighed (Seca 877; Seca GmbH & Co, Hamburg, Germany). They then prepared for the exhaustive bouts of exercise starting at 1000 h. The training session was initiated with an incremental fat oxidation profile warm-up followed by 2 min recovery and a 10-s maximal sprint (Sprint 1). After 4 min of recovery, participants cycled at $W_{68\%}$ until exhaustion. Exhaustion was defined as when participants could not maintain their pedal frequency over 60 rpm at the desired work rate despite three verbal encouragements to maintain their pedaling frequency. After exhaustion, participants recovered for 5 min and then performed 1-min intervals at a power output eliciting 90% of $\dot{V}O_{2max}$ ($W_{90\%}$) followed by 1 min recovery, repeatedly, until voluntary exhaustion. $\dot{V}O_2$ and RER were measured continuously during the warm-up, every 15 min, and at exhaustion during the exercise at $W_{68\%}$ and continuously during the intervals at $W_{90\%}$ for calculation of carbohydrate and fat oxidations. Lactate, glucose, other metabolites, and blood gas parameters were measured (ABL800; Radiometer Medical, Copenhagen, Denmark). HR was measured continuously (Polar RS800CX, Kempele, Finland).

During the exhaustive exercise, participants consumed flavored water the first time and a carbohydrate drink (1 $g\cdot kg^{-1}\cdot h^{-1}$) the second time. Drinks were served in opaque flasks and were indistinguishable by taste. The drinks were supplied in equal portions every 15 min. The carbohydrate drink was a 10% solution containing 50% glucose (GPR RECTAPUR®; VWR Chemicals, Radnor, PA) and 50% maltodextrin (Dextri-maltose®; MP Biomedicals, Santa Ana, CA). All drinks contained the same quantity of electrolytes

TABLE 1. Energy intake and content of macronutrients for the different standardized meals and supplements during the study protocol.

	Carbohydrate ($g\cdot kg^{-1}$)	Protein ($g\cdot kg^{-1}$)	Fat ($g\cdot kg^{-1}$)	Energy Content ($kJ\cdot kg^{-1}$)
Dinner the day before (~9:00 PM)	2.03	0.81	0.57	69.68
Day 1				
Exhaustive training protocol drink (~10:00 AM) (CHO/PLA)	2.50/0	0.00	0.00	42.50/0
Postexercise drink (~2:00 PM)	1.20	0.00	0.00	20.40
Postexercise meal (~2:30 PM)	2.02	0.48	0.07	45.03
Afternoon meal (~4:00 PM)	2.51	0.50	0.21	59.10
Evening drink (~5:00 PM) (CHO/PLA)	0/2.50	0.00	0.00	0/42.50
Dinner (~7:00 PM)	2.02	0.68	0.12	50.61
Day 2				
Postexercise drink (~9:00 AM)	1.20	0.00	0.00	20.40
Total of days 1 and 2	11.44	1.67	0.40	238.00
Absolute total	892 g	130 g	31 g	18,564 kJ

The two conditions were isoenergetic between groups and adjusted for body weight. The exercise started at 10:00 am on day 1 and 7:00 am on day 2

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(0.7 g·L^{-1} NaCl) and were flavored with 100 g·L^{-1} of drink flavoring (Fun light, Stabburet, Norway). In total, participants ingested 193 ± 21 g of carbohydrates during $W_{68\%}$.

During the intervention, all participants were free to drink additional water. The total fluid intake was 3256 ± 208 mL during CHO and 3340 ± 274 mL during PLA, with no significant difference between conditions (P = 0.342). The total weight loss during exercise was lower in CHO than in PLA (CHO, 3.2 ± 0.3 kg; PLA, 3.6 ± 0.3 kg; P = 0.048).

The aim of the present study was to compare an exercise of cycling to exhaustion without energy supply versus the same duration of exercise in which participants consumed a carbohydrate drink to prevent exhaustion. As the interindividual variation in time to exhaustion varies greatly whereas intraindividual variation is much less (16), the design of the present investigation was inspired by the study of Coyle et al. (12). In this design, participants received the placebo at the first test to determine the time points for measurements during the carbohydrate supplementation condition. In the present study, participants were blinded (single-blinded design) and exercised until exhaustion under the placebo condition at the first test to determine the quantity of work. In the second test, carbohydrate was supplied, and participants completed exactly the same amount of work before being stopped by the experimenter. The two interventions were separated by at least 7 d to allow participants to recover (17–19).

After the exercise, participants received a drink with carbohydrate (1.2 g carbohydrate·kg^{-1}; 17%) that was to be consumed within 5 min. The last blood sample was collected 30 min after ingestion of this drink. After this blood sample and once they had provided the first urine sample after exercise, participants were served a standardized meal (composed of pasta, tomato sauce, chocolate protein pudding, and a banana). This meal had a macronutrient content of carbohydrate 2.02 g·kg^{-1}, protein 0.48 g·kg^{-1}, and fat 0.07 g·kg^{-1} (45.0 kJ·kg^{-1}). The participants returned home with an additional drink and evening food. Participants were asked to consume a meal at 1600 h composed of oats, raisins, milk, and a banana. The macronutrient content of this meal was carbohydrate 2.51 g·kg^{-1}, protein 0.50 g·kg^{-1}, and fat 0.21 g·kg^{-1} (59.1 kJ·kg^{-1}). During PLA, participants consumed a drink containing 1 g carbohydrate·kg^{-1} at 1700 h, whereas during CHO they received a similar-flavored water drink without energy. This was done to match the energy content of the carbohydrate drink consumed during the exhaustive exercise in CHO and avoid any energy difference between the two conditions that could explain differences in performance in the time trial test the next day (14). At 1900 h, participants had a prepared dinner (composed of couscous, kidney beans, corn, chopped tomatoes, milk, and an apple) with the following energy content: carbohydrate 2.02 g·kg^{-1}, protein 0.68 g·kg^{-1}, and fat 0.12 g·kg^{-1} (50.6 kJ·kg^{-1}). Water was allowed *ad libitum*. Participants were not allowed to eat food other than that provided.

Day 2. The following day, participants reported to the laboratory at 0630 h after an overnight fast. If the participants lived further than 2 km from the laboratory, they were instructed to come by car or public transportation. A Venflon catheter was inserted into an antecubital vein. Participants were weighed and then prepared for the performance test. The performance test was defined as a time trial cycling exercise. At 0700 h, approximately 17 h after the end of the exhaustive exercise, participants completed the incremental fat oxidation profile warm-up. After 2 min recovery, a 10-s maximal sprint was performed (sprint 2). After that, participants cycled 15 min at $W_{68\%}$. After 5 min of recovery, participants started the time trial TT, which was the same as during the familiarization day. Participants were blinded to time during TT and received only feedback on the accumulated external energy produced (percent from target total work output). They did not receive their TT performance times until they had completed both TT. Two minutes after TT, participants performed a 10-s maximal sprint (sprint 3).

After sprint 3, participants received a drink with carbohydrate (1.2 g carbohydrate·kg^{-1}; 17%). The study was designed with similar total energy intakes in CHO and PLA during the protocol (Table 1). On days 1 and 2, there were mean intakes of carbohydrate 11.44 g·kg^{-1}, protein 1.67 g·kg^{-1}, and fat 0.40 g·kg^{-1} (238.0 kJ·kg^{-1}). Over 24 h, this meant intakes of carbohydrate 13.08 g·kg^{-1}, protein 1.90 g·kg^{-1}, and fat 0.46 g·kg^{-1} (272.1 kJ·kg^{-1}).

$\dot{V}O_2$ and RER were measured continuously during the incremental fat oxidation profile warm-up, at the start and the end of the exercise at $W_{68\%}$, at the start of TT, as the rider completed 20%, 40%, 60%, 80%, and continuously during the last 5% of TT. HR was recorded continuously. Participants reported a rate of perceived exertion according to the Borg scale. Gross efficiency was calculated during TT as [gross efficiency (%) = work accomplished (kJ)/energy expended (kJ) × 100]. The energy expended was calculated using the $\dot{V}O_2$ and RER measured continuously during TT (20).

Biological Samples

Blood analyses. Venous blood samples were drawn anaerobically with a 1-mL heparinized syringe (SafePICO, Radiometer, Denmark) for blood gas parameter analysis. Venous blood samples were also drawn anaerobically with 10-mL heparinized syringes (Omnifix, B Braun, Melsungen, Germany), into 6-mL K2E K2EDTA tubes (Vacuette tubes; Greiner Bio-One, Kremsmünster, Austria) and into 5-mL lithium heparin tubes (Vacuette tubes, Greiner Bio-One) and placed on ice. Blood samples were collected before and at the end of the warm-up, before the exhaustive exercise ($W_{68\%}$), and every 30 min during the exercise at $W_{68\%}$, at exhaustion at $W_{68\%}$, at exhaustion at $W_{90\%}$, and 30 min after all exercise (day 1). Blood samples were also collected before and at the end of the warm-up; before and at the end of the exercise at $W_{68\%}$; before, and at the 25%, 50%, 75% stages, and at the end of TT; and 30 min after all exercise (day 2).

All blood samples were kept on ice until centrifugation (2500 g; 10 min; 4°C), pipetted into 1.5-mL Eppendorf tubes, and stored at −80°C until subsequent analysis. All analytic processes were conducted according to the manufacturer's instructions.

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Blood gas analyzer. Blood gas parameters and blood metabolite concentrations were determined with an ABL800 (Radiometer, Copenhagen, Denmark). Standard plasma bicarbonate, equilibrated to CO_2 and O_2 partial pressures of 40 and 100 mm Hg, respectively, was calculated according to the ABL800 instruction manual.

Plasma amino acids. Plasma amino acid (AA) concentrations were determined from blood samples collected into EDTA tubes at the start of $W_{68\%}$ and at the end of $W_{90\%}$ to quantify the effect of the exhaustive exercise on the amino acid plasma levels.

Plasma AA were quantified using liquid chromatography–tandem mass spectrometry (LC-MS/MS). Briefly, isotopically labeled internal standards were added to plasma, followed by reduction of disulfides using dithioerythritol and then protein precipitation using 5-sulfosalicyclic acid. The extracts were diluted with an aqueous solution of formic acid (0.5%) and heptafluorobutyric acid (HFBA) (0.3%) before analysis. LC-MS/MS was carried out using a Shimadzu LC-20ADXR Prominence LC system (Kyoto, Japan) coupled to a Sciex QTRAP5500 mass spectrometer with a Turbo V ion source and TurboIonspray probe (Framingham, MA). Chromatographic separation was achieved on a Phenomenex Kinetex Core Shell C18 (100 × 4.6 mm, 2.6 μm) LC column (Torrance, CA) with an aqueous solution of formic acid (0.5%) and HFBA acid (0.3%) and acetonitrile gradient mobile phase. Positive mode multiple reaction monitoring was used for detection. Linear calibration curves of the peak area ratios of analyte and internal standard were used for quantification. The coefficients of variation for the analytes were 3.4% and 6.7%.

Plasma ammonia. Plasma ammonia was quantified from blood samples collected into EDTA tubes at the start of $W_{68\%}$ and the end of $W_{90\%}$ using an enzymatic UV kit (AM 1015; Randox Laboratories Ltd., Crumlin, United Kingdom).

Urine analyses. All the urine samples from the participants were collected in plastic containers in four consecutive batches: 1) from the start of day 1 until the start of exhaustive exercise (duration ~10 h), 2) from after the start of exhaustive exercise until the first urine after exhaustive exercise (~4 h), 3) from the first urine after exhaustive exercise until midnight of day 1 (~10 h), and 4) from midnight of day 1 until the start of TT (~7 h). Participants were instructed to void their bladders before the start of the exhaustive exercise. Volume was measured for each period, and a 15-mL sample was frozen at −20°C for analysis.

Urinary 3-methylhistidine. The urines from periods 2, 3, and 4 were mixed in proportions relative to the volume of urine during these periods. The concentration of 3-methylhistine was then quantified in these pooled urine samples using LC-MS/MS.

Calculation of nitrogen balance. The nitrogen concentration in the urine was analyzed using the Kjeldahl method (21). The nitrogen balance was calculated based on protein intake and urinary nitrogen excretion assuming that nitrogen makes up 16% of the protein and a nonurinary nitrogen loss of 23.2% (22). These calculations were carried out over a period of 21 h for the period of the protocol after the consumption of the drinks. Nitrogen intake was calculated assuming a nitrogen-to-amino acid constant of 6.25 (23).

Calculation of Carbohydrate and Fat Oxidation

Carbohydrate and fat oxidation were calculated from RER, oxygen uptake, and nitrogen excretion, according to the method of Jeukendrup and Wallis (24).

RPE

RPE was rated on the 6–20 Borg scale (25).

Statistics

All values are expressed as mean ± SEM. Shapiro–Wilk tests were carried out to investigate whether the data were normally distributed: the data passed these tests, and parametric tests were used. Two-way repeated-measures ANOVA were used to determine treatment differences, with treatment and time used as factors. When differences were detected, Tukey's honest significant difference *post hoc* test was performed. Performances during TT were compared using a paired sample *t*-test and calculating Cohen's *d* effect size by dividing their difference score by their pooled SD. Differences were considered significant when $P < 0.05$. Statistical analyses were performed using Sigma Plot 14.0 software.

RESULTS

Exhaustive Exercise Day 1

Participants cycled to exhaustion at a relative intensity of 68.2% ± 0.7% of $\dot{V}O_{2max}$, corresponding to a work rate of 236 ± 10 W ($W_{68\%}$). Participants cycled for 147 ± 17 min (range, 95 to 225 min) before voluntary exhaustion in PLA. After 5 min of recovery, they performed 8 ± 1 1-min intervals at 329 ± 18 W corresponding to 90% of $\dot{V}O_{2max}$ ($W_{90\%}$). Participants completed the same amount of exercise a week later while receiving CHO.

Indirect Calorimetry

RER was significantly higher in CHO compared with PLA during $W_{68\%}$ (CHO, 0.918 ± 0.004; PLA, 0.882 ± 0.005; $P = 0.030$) (Fig. 2B), resulting in larger total carbohydrate oxidation ($P = 0.018$) (Table 2) and lower total fat oxidation in CHO than in PLA (CHO, 82 ± 12 g; PLA, 128 ± 25 g; $P = 0.028$). The protein oxidation was not significantly different between CHO and PLA (CHO, 25 ± 5 g; PLA, 28 ± 2 g; $P = 0.599$).

$\dot{V}O_2$, ventilation, HR, and RPE increased during $W_{68\%}$ (time effect, $P < 0.001$), but with no significant difference between conditions (Fig. 2A, E, F, G).

Plasma Metabolites

The plasma glucose concentration continually decreased from 5.2 ± 0.2 mM at baseline to 3.7 ± 0.4 mM at exhaustion in PLA ($P < 0.001$) (Fig. 2H). The plasma glucose concentration

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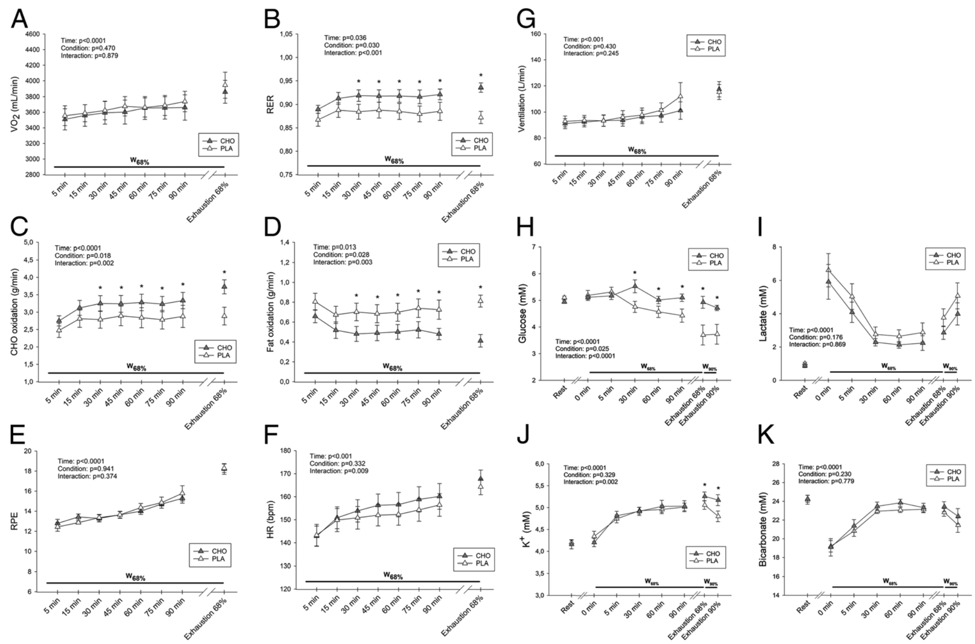

FIGURE 2—Metabolic and blood parameters during the exhaustive exercise $W_{68\%}$ on day 1. A, Absolute oxygen consumption. B, RER. C, Absolute carbohydrate oxidation. D, Absolute fat oxidation. E, Rate of perceived exertion. F, HR. G, Ventilation. H, Venous plasma glucose concentration (mM). I, Venous plasma lactate concentration (mM). Note that venous plasma lactate concentrations are high at the start of exercise due to the maximum sprint performed 2 min before the start of exercise until exhaustion. J, Venous plasma potassium K^+ concentration (mM). K, Standard venous plasma bicarbonate concentration (mM). Values are presented as mean ± SEM. *P < 0.05, time points with differences between carbohydrate (CHO) and placebo (PLA).

did not change during the exhaustive exercise in CHO (P = 0.523): from 5.1 ± 0.1 mM at baseline of $W_{68\%}$ to 4.7 ± 0.1 mM at the end of $W_{90\%}$, leading to a significant interaction effect (time–condition, P < 0.0001).

Participants performed a 10-s maximal sprint 2 min before $W_{68\%}$ started. Therefore, the plasma lactate concentration was high at the start of $W_{68\%}$. In CHO and PLA, plasma lactate concentrations did not differ significantly (P = 0.176) during the exhaustive exercise (Fig. 2I).

The plasma potassium concentration increased continually during exercise in both CHO and PLA (time effect P < 0.0001). At the end of $W_{90\%}$, plasma concentrations of potassium were significantly higher in CHO than in PLA (P = 0.005) (Fig. 2J).

The plasma bicarbonate concentration decreased during the incremental fat oxidation profile warm-up and the initial sprint before increasing continually during $W_{68\%}$ toward the rested value in both CHO and PLA (time effect P < 0.0001) (Fig. 2K).

The plasma ammonia concentration was not different before exercise and at exhaustion between CHO and PLA. There was also no significant effect of the exhaustive exercise between the two conditions.

Postexercise Recovery Drink

Post-$W_{90\%}$, the participants consumed a recovery carbohydrate drink under both conditions. Plasma glucose rose in both CHO and PLA (CHO, from 4.7 ± 0.1 mM after $W_{90\%}$ to 6.6 ± 0.3 mM 30 min after the consumption of the recovery drink; PLA, from 3.7 ± 0.4 mM to 6.3 ± 0.5 mM; time effect P < 0.001). There was no significant difference in the increase between the conditions (P = 0.541). Moreover, plasma lactate concentrations decreased 30 min after the consumption of the recovery drink, with no difference between conditions (P = 0.600).

Performance Day 2

Incremental fat oxidation profile warm-up. During the initial incremental fat oxidation profile warm-up on day 2, fat oxidation (P = 0.066) tended to be higher on day 2 after PLA than after CHO. Maximal fat oxidation was 0.61 g·min^{-1} after PLA and 0.55 g·min^{-1} after CHO (P = 0.149). Fat_{max} was 39.2% $\dot{V}O_{2max}$ after PLA and 36.5% $\dot{V}O_{2max}$ after CHO (P = 0.264). $\dot{V}O_2$, HR, and RPE increased with exercise intensity (P < 0.001) but with no significant difference between conditions (Supplemental Fig. 1, Supplemental Digital Content, Metabolic and blood parameters during the incremental fat oxidation profile warm-up on day 2, http://links.lww.com/MSS/C894).

Sprint 2. During the following 10-s maximal sprint (sprint 2), mean power output was significantly higher after CHO than after PLA (CHO, 1024 ± 62 W; PLA, 951 ± 86 W; P = 0.011) (Fig. 3C) and decreased from day 1 (sprint 1) only

TABLE 2. Carbohydrate oxidation during exercise after consumption of the drinks on day 1.

Carbohydrate Oxidation (g)	CHO	PLA	P
$W_{68\%}$	467 ± 50	386 ± 30	0.020
$W_{90\%}$	50 ± 7	40 ± 5	0.047
Total	518 ± 55	426 ± 31	0.022

Values are obtained from indirect calorimetry measurements. Values are presented as mean ± SEM.

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in PLA (P = 0.009). During sprint 1 on day 1, there was no significant difference in mean power output between the two conditions (CHO, 1050 ± 59 W; PLA, 1060 ± 71 W; P = 0.716).

Pre-TT. After the sprint, participants cycled for 15 min at the same power output of $W_{68\%}$ as the exhaustive exercise on day 1 (Pre-TT). Ventilation (P = 0.018), HR (P = 0.020), and RPE (P = 0.002) were higher on day 2 than during the first 15 min on day 1 (Pre-TT vs $W_{68\%}$).

Gross efficiency decreased during $W_{68\%}$ on day 1 in both PLA and CHO, with no difference between the two conditions (P = 0.708). During Pre-TT, gross efficiency returned to the baseline level at the start of $W_{68\%}$ on day 1 after both CHO and PLA (Fig. 3D).

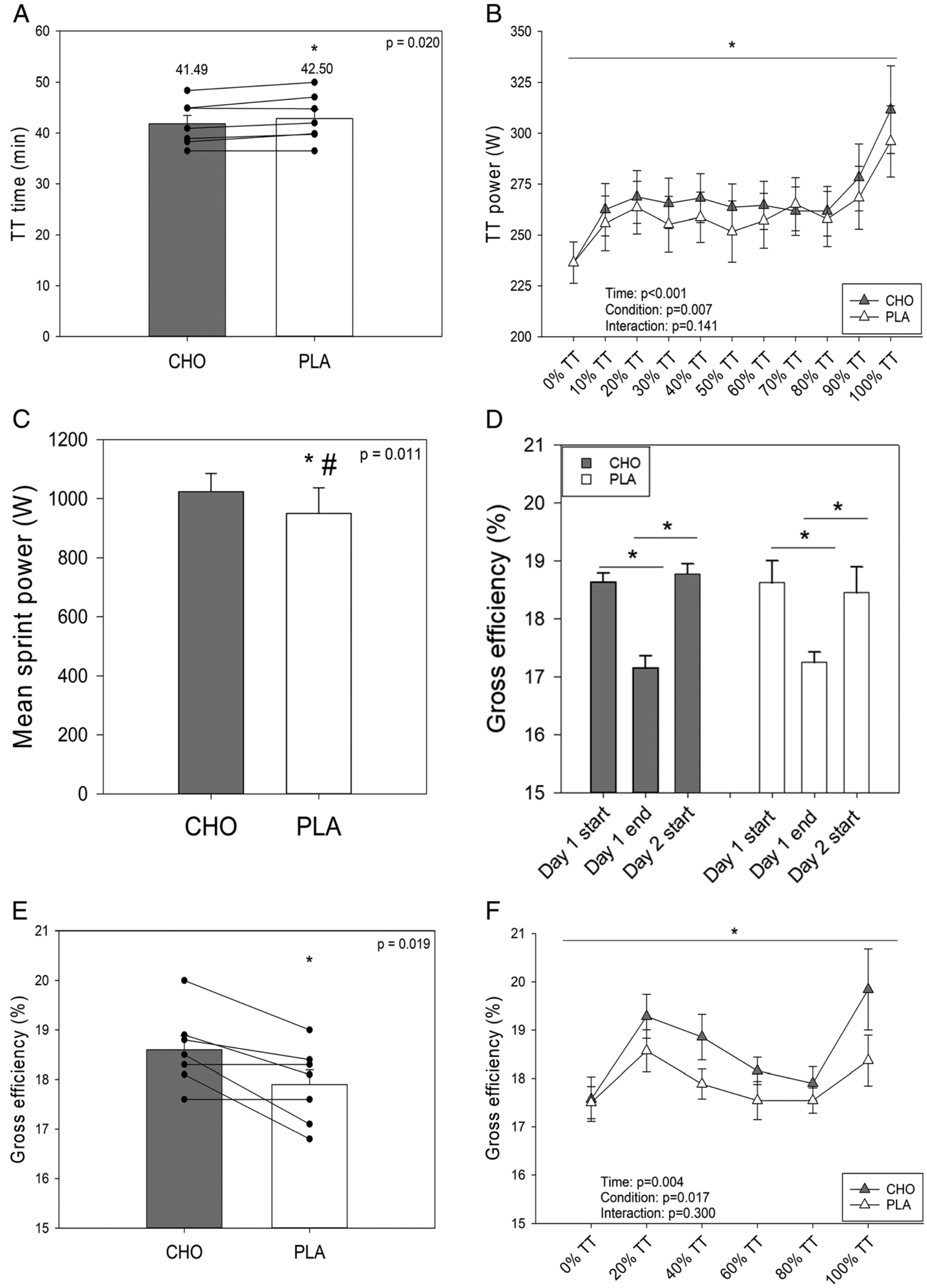

FIGURE 3—Performance and power output during the time trial (TT). A, Average time spent on TT. B, Mean power output curve during TT. During TT, the participants had to complete a specific amount of mechanical work. They could adjust the work rate on a panel mounted on the bike as they preferred. C, Mean power output during the 10-s sprint test sprint 2, on day 2 before $W_{68\%}$. D, Evolution of gross efficiency between the start of $W_{68\%}$ on day 1, the end of $W_{68\%}$ on day 1, and the end of Pre-TT on day 2. E, Average gross efficiency during TT. F, Gross efficiency during TT. Values are presented as mean ± SEM. *P < 0.05, compared with carbohydrate (CHO). #P < 0.05, compared with sprint 1.

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$\dot{V}O_2$, RER, carbohydrate oxidation, fat oxidation, ventilation, HR, and RPE did not significantly differ between conditions during Pre-TT. Plasma glucose and lactate decreased significantly during Pre-TT (glucose, $P = 0.005$; lactate, $P = 0.005$) with no difference between conditions (glucose, $P = 0.144$; lactate, $P = 0.101$) (Fig. 4H, I). Plasma glucose tended to be lower at the start of Pre-TT than of $W_{68\%}$ on day 1 ($P = 0.052$), but not significantly after CHO compared with after PLA (CHO, day 1 = 5.2 ± 0.1 mM, day 2 = 4.9 ± 0.1 mM, $P = 0.112$; PLA, day 1 = 5.3 ± 0.2 mM, day 2 = 5.0 ± 0.1 mM; interaction $P = 0.153$).

Time trial. After CHO, participants performed TT 2.4% faster than after PLA (CHO, 41:49 ± 1:38 min; PLA, 42:50 ± 1:46 min; $P = 0.020$; effect size $d = 0.23$, small) (Fig. 3A), with a significantly higher mean power output (CHO, 268 ± 4 W; PLA, 261 ± 4 W; $P = 0.007$) (Fig. 3B).

Gross efficiency during TT was significantly higher on average after CHO than after PLA ($P = 0.019$) (Fig. 3E). Starting from the same level of gross efficiency at the start of TT, participants maintained better gross efficiency during the remainder of TT after CHO (Fig. 3F).

Furthermore, after PLA, TT performance was significantly lower than during the familiarization session ($P = 0.030$) but not after CHO ($P = 0.058$). The familiarization session was performed rested and was completed in 39:14 ± 0:55 min with a mean power output of 281 ± 5 W.

$\dot{V}O_2$, RER, carbohydrate oxidation, fat oxidation, HR, RPE, and ventilation increased during TT ($P < 0.001$), but with no significant difference between the two conditions, although after CHO the participants cycled at a significantly higher power output (Fig. 4A, B, C, D, E, F, G).

Plasma glucose decreased during the start of TT before increasing continually toward the end (time effect, $P < 0.0001$) (Fig. 4H). Plasma glucose was significantly higher after PLA than after CHO ($P = 0.005$). There were no significant differences between interventions during TT concerning plasma concentrations of lactate ($P = 0.078$) (Fig. 4I). Only the lactate concentration at the end of TT was significantly higher after PLA than after CHO ($P = 0.002$).

Sprint 3. The participants finished the performance session on day 2 with a second 10-s maximal sprint 2 min after TT (sprint 3). The mean power output was not significantly different between conditions (CHO, 996 ± 76 W; PLA, 951 ± 67 W; $P = 0.103$). However, it should be noted that mean power output after PLA appeared significantly lower during sprint 3 than sprint 1 ($P = 0.009$), but there was no significant difference after CHO ($P = 0.277$).

Nitrogen balance and urinary nitrogen excretion. During the 21-h study period, nitrogen intake was 266.5 ± 3.7 $mg \cdot kg^{-1}$ (20.7 ± 0.7 g), assuming that nitrogen makes up 16% of the protein (in mass) (23). Relative total nitrogen excretions during this period were 232.3 ± 15.2 $mg \cdot kg^{-1}$ for CHO and 234.5 ± 11.6 $mg \cdot kg^{-1}$ for PLA (Fig. 5A) and were not significantly different ($P = 0.897$). Moreover, the nitrogen balance in the two conditions was positive (CHO, 34.2 ± 14.6 $mg \cdot kg^{-1}$; PLA, 32.0 ± 11.1 $mg \cdot kg^{-1}$), and there was no significant difference between the two conditions ($P = 0.803$).

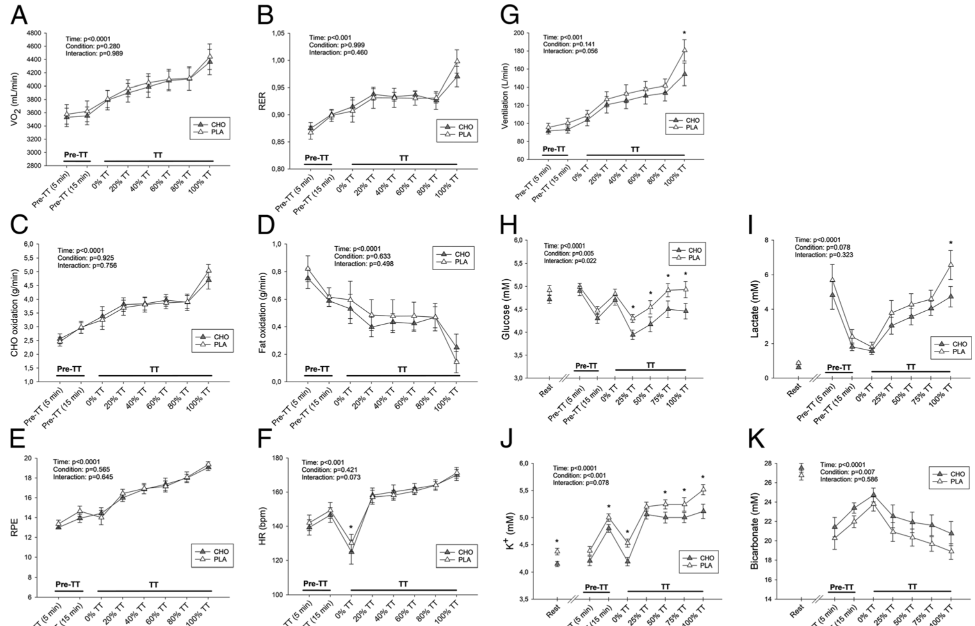

FIGURE 4—Metabolic and blood parameters during Pre-TT and TT on day 2. A, Absolute oxygen consumption. B, RER. C, Absolute carbohydrate oxidation. D, Absolute fat oxidation. E, Rate of perceived exertion. F, HR. G, Ventilation. H, Venous plasma glucose concentration (mM). I, Venous plasma lactate concentration (mM). Note that venous plasma lactate concentrations are high at the start of exercise due to the maximum sprint completed 2 min before the start of Pre-TT. J, Venous plasma potassium K^+ concentration (mM). K, Venous plasma bicarbonate concentration (mM). Values are presented as mean ± SEM. *$P < 0.05$, time points with differences between carbohydrate (CHO) and placebo (PLA).

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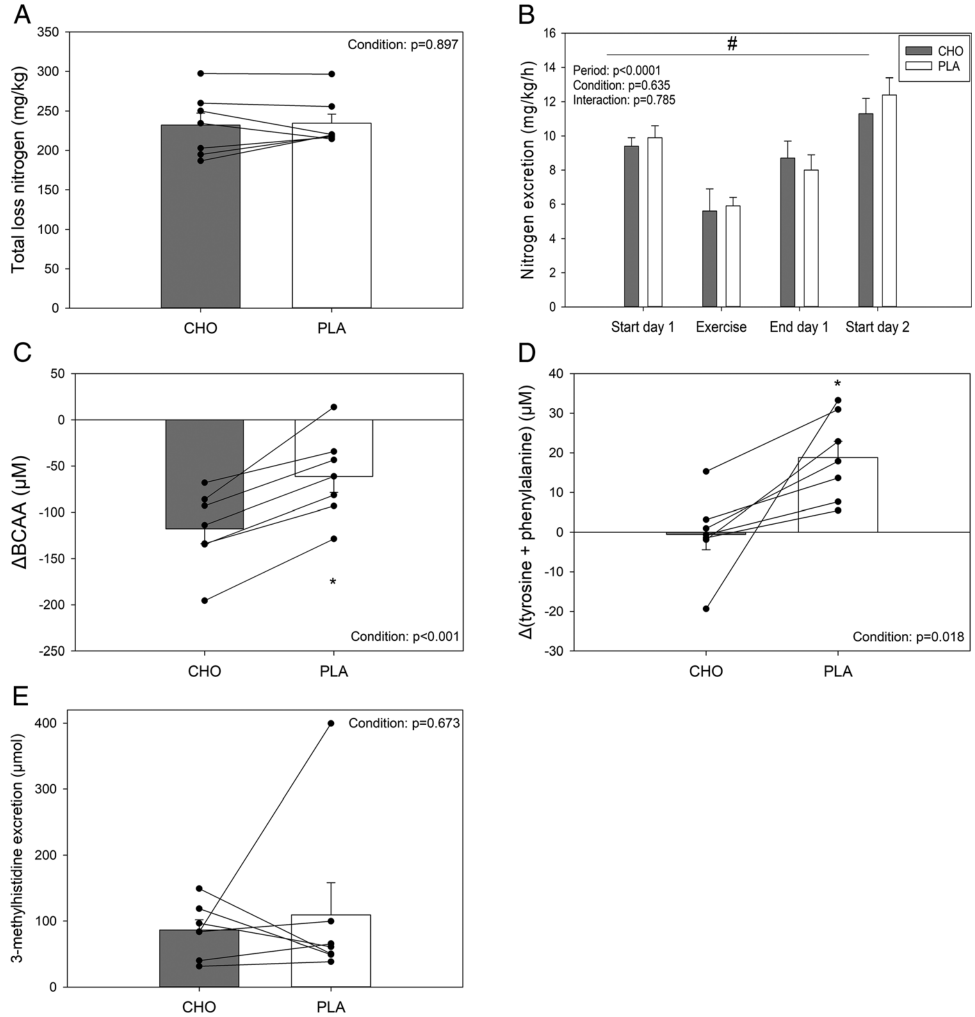

FIGURE 5—Urine nitrogen concentration during the 21-h study protocol period and difference in plasma amino acid concentrations before and after exhaustive exercise A, Relative levels of total urine nitrogen. B, Relative levels of total urine nitrogen through the various periods adjusted for time. C, Difference in plasma BCAA concentrations during exhaustive exercise. D, Difference in plasma tyrosine and phenylalanine concentrations during exhaustive exercise. E, 3-Methylhistidine excretion. The differences in plasma amino acid concentrations were calculated as the concentration after exercise minus the concentration before exercise. A negative value indicates that the concentration after is less than the value before. Values are presented as mean ± SEM. ****P* **< 0.05, compared with carbohydrate (CHO). #*P* < 0.05, the effect of the time.

Urine nitrogen excretion, normalized for time, did not significantly differ between the two conditions ($P = 0.635$; start day 1, $P = 0.732$; exercise, $P = 0.713$; end day 1, $P = 0.511$; start day 2, $P = 0.345$) (Fig. 5B). However, there was a significant effect of time ($P < 0.0001$) on the relative total nitrogen excretion: these quantities were significantly higher at the start of day 2 than during exercise ($P < 0.001$) and at the end of day 1 ($P = 0.002$) as well as at the start of day 1 compared with during exercise ($P = 0.006$).

Plasma amino acid concentrations on day 1. Plasma amino acid concentrations were measured before exercise and at exhaustion on day 1 (Table 3 and Supplemental Table 1, Supplemental Digital Content, Plasma amino acid concentrations before and after exhaustive exercise in PLA and CHO, http://links.lww.com/MSS/C894).

The plasma concentrations of the AA tyrosine and phenylalanine increased from before to after exhaustive exercise ($W_{68\%}$ and $W_{90\%}$) only in PLA (tyrosine, the difference from zero $P = 0.025$; phenylalanine, $P = 0.008$) and remained stable in CHO (tyrosine, $P = 0.483$; phenylalanine, $P = 0.677$) (Table 3). Moreover, plasma tyrosine ($P = 0.033$) and phenylalanine ($P = 0.026$) concentrations increased more in PLA than in CHO (Fig. 5D).

The plasma concentrations of branched-chain amino acid (BCAA) decreased at exhaustion in both conditions (Table 3), but more in CHO than in PLA ($P < 0.001$) (Fig. 5C).

3-Methylhistidine excretion. 3-Methylhistidine excretion in the pooled urine samples was lower in CHO (86.3 ± 15.6 μmol) than in PLA (109.3 ± 48.9 μmol) but was not significantly different between the two conditions ($P = 0.673$) (Fig. 5E).

DISCUSSION

Time trial performance was ~1 min (2.4%) better the day after ~2.5 h of fatiguing exercise while consuming a carbohydrate

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TABLE 3. The change in plasma amino acid concentrations (μM) from before to after the exhaustive exercise ($W_{68\%}$) in the PLA or CHO trials.

	ΔPLA (μM)	ΔCHO (μM)	Difference Trials (*P*-Value)
Phenylalanine	10.6 ± 2.7	1.0 ± 2.2*	**0.026**
Tyrosine	8.2 ± 2.8	−1.5 ± 2.1*	**0.033**
Leucine	−16.6 ± 4.2	−36.5 ± 4.1*	***P* < 0.001**
Isoleucine	−7.1 ± 2.7	−17.3 ± 3.7*	**0.017**
Valine	−37.5 ± 11.9	−64.0 ± 8.9*	**0.009**
BCAA	−61.2 ± 17.4	−117.8 ± 16.0*	***P* < 0.001**
Glutamic acid	9.1 ± 4.0	22.5 ± 3.4*	**0.004**
Aspartic acid	0.1 ± 0.5	1.1 ± 0.5*	**0.045**
Alanine	−66.5 ± 41.8	−77.5 ± 18.3	0.777
Glutamine	−42.7 ± 42.3	−47.5 ± 22.3	0.912
Arginine	−14.4 ± 7.3	−13.4 ± 3.6	0.846
Asparagine	−3.9 ± 5.7	−6.5 ± 3.0	0.599
Cysteine	29.9 ± 8.6	33.6 ± 4.9	0.720
Glycine	−35.8 ± 16.2	−39.2 ± 6.7	0.814
Histidine	−4.4 ± 7.6	−6.3 ± 2.4	0.778
Lysine	−28.4 ± 11.4	−26.1 ± 8.2	0.779
Methionine	1.6 ± 1.4	−0.2 ± 0.8	0.304
Proline	−53.7 ± 15.1	−61.8 ± 12.4	0.362
Serine	−9.3 ± 6.0	−22.9 ± 4.2	0.094
Threonine	−18.9 ± 10.2	−16.8 ± 4.7	0.811
Tryptophan	4.2 ± 1.6	2.6 ± 1.6	0.575
Creatinine	21.4 ± 6.2	27.1 ± 5.4	0.410
Cystathionine	0.3 ± 0.1	0.2 ± 0.0	0.094
Glutathione	1.5 ± 0.4	1.8 ± 0.3	0.547
Homocysteine	0.7 ± 0.4	1.2 ± 0.3	0.343
Ornithine	−11.9 ± 2.8	−14.4 ± 2.4	0.210
Taurine	70.5 ± 18.8	92.2 ± 11.6	0.116

The raw values before and after the exhaustive exercise can be obtained in the supplements (Supplemental Table 1, Supplemental Digital Content, Plasma amino acid concentrations before and after exhaustive exercise in PLA and CHO, http://links.lww.com/MSS/C894). The BCAA concentration is the sum of the concentrations of leucine, isoleucine, and valine. Values are presented as mean ± SEM. *$P < 0.05$, compared with placebo (PLA).

drink compared with after the same exercise carried out while consuming a similar-flavored placebo drink. Nitrogen- and 3-methylhistidine excretion were similar in both conditions, which argues against our hypothesis that carbohydrate intake would reduce protein degradation. However, this hypothesis is supported by the higher tyrosine and phenylalanine plasma concentrations at exhaustion and a lower gross efficiency during the time trial in PLA than in CHO.

Carbohydrate intake maintained carbohydrate availability. Inspired by Coyle et al. (12), the present experimental study was designed for participants to reach exhaustion in PLA on day 1. By contrast, ingestion of carbohydrate would prevent exhaustion and reduce metabolic stress. It is well-documented that carbohydrate intake during exercise prolongs time to exhaustion (12–14,26,27). As in Coyle et al. (12), there was no significant difference between the conditions during exercise on day 1 in RPE, oxygen uptake, and HR. Plasma lactate concentration tended to be higher in PLA, as Coyle et al. (12) also reported, but the prevention of the development of exhaustion was mainly induced by the maintenance of carbohydrate availability.

Ingesting carbohydrates during exercise significantly increased carbohydrate oxidation during exercise (518 ± 55 g in CHO vs 426 ± 31 g in PLA) and supports the findings of Coyle et al. (12). The carbohydrate intake in CHO was 193 ± 21 g, leading to a net utilization of endogenous glycogen of 325 ± 34 g in CHO. This was significantly lower than the net carbohydrate oxidation in PLA from endogenous reserves. However, Coyle et al. (12) did not report any difference in glycogen utilization between cycling to exhaustion with or without carbohydrate ingestion: the increase in net carbohydrate oxidation came from greater glucose availability in the blood. In the present study, carbohydrate intake during exercise prevented the decline in plasma glucose observed in PLA. Participants did not reach hypoglycemia at exhaustion in CHO, in agreement with Coyle et al. (12). Similarly, plasma glucose was maintained at resting levels during the whole cycling exercise in CHO. All this indicates that ingesting carbohydrates during exercise maintained carbohydrate availability.

Protein degradation measured with urinary nitrogen excretion. In the present study, nitrogen excretion was not different between conditions. Our hypothesis was that ingesting carbohydrates during exercise would reduce urinary nitrogen excretion. Indeed previous studies have shown greater protein degradation when endogenous carbohydrate availability was limited (6,8). Based on our data, carbohydrate availability was maintained during exercise in CHO, but this did not lead to a difference in urinary nitrogen excretion.

Rustad et al. (9) reported in a double-blinded, randomized, balanced crossover design with exhaustive exercises on two consecutive days that nitrogen excretion was lower when carbohydrate was ingested in recovery (PLA, 195.8 ± 13.6 $mg \cdot kg^{-1}$; CHO, 169.7 ± 9.3 $mg \cdot kg^{-1}$; $P = 0.05$). Similarly, the ingestion of 100 g of carbohydrate 1 h after a resistance exercise bout (10 sets of 8 repetitions of leg presses at 80% of 1RM) resulted in decreased muscle protein breakdown compared with the placebo (28). Nitrogen excretion provides a reasonable estimate of whole-body protein degradation (29): a parallel can be drawn between nitrogen excretion and protein degradation. This suggests that a lower energy and/or carbohydrate availability increased nitrogen excretion in the above studies. In the present study, the diet was standardized and the energy intake was exactly similar between the conditions as well as the carbohydrate intake. Only the timing of the intake differed.

Differences in nitrogen excretion in other compartments, such as sweat and feces, may explain the absence of difference in nitrogen excretion. Lemon and Mullin (6) reported higher sweat nitrogen concentrations during exercise with low carbohydrate availability than with high. This was not considered in the present study, in which, due to practical considerations, nitrogen excretion in sweat was estimated. A measurement of sweat nitrogen excretion could have led to differences between the conditions, especially because the sweat rate was significantly higher in PLA than in CHO ($P = 0.049$).

Protein degradation measured with urinary 3-methylhistidine excretion. 3-Methylhistidine constitutes an integral part of the myofibrillar proteins found in both actin and myosin (15). It is neither reutilized for protein synthesis (30) nor metabolized (31) and is excreted in the urine. Consequently, the urinary excretion of 3-methylhistidine may be used to estimate the rate of myofibrillar protein breakdown (32,33). In the present study, participants consumed a 3-methylhistidine-free diet for the last 3 d before each intervention's start and during the study protocol. An increase in 3-methylhistidine excretion would thus mean an increase in myofibrillar protein breakdown.

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Urinary 3-methylhistidine excretion was not significantly different between the conditions. Our data do not support the idea that carbohydrate intake reduces degradation in the myosin and actin muscles' myofibrillar protein compartment.

In the present study, daily urinary excretion of 3-methylhistidine was 98.6 ± 17.8 μmol·day^{-1} in CHO and 124.9 ± 55.1 μmol·day^{-1} in PLA. Both were lower than the mean daily urinary excretion of 3-methylhistidine in a similar meat-free diet, which has been reported to be ~200 μmol·day^{-1} (34,35). This could be explained by the fact that the urinary production of 3-methylhistidine is said to fall during exercise (36). Similarly, Carraro et al. (37) did not show an increase in 3-methylhistidine excretion during an exercise versus a control period. However, excretion was significantly elevated in the samples after the exercise period. Measuring 3-methylhistidine excretion during the days after the exhaustive exercise could resolve this issue. Including a control day without exercise in this study could also have made it possible to measure the basal level of urinary excretion of 3-methylhistidine and thus helped to interpret the difference between the conditions.

Protein degradation measured with tyrosine and phenylalanine. Plasma tyrosine and phenylalanine concentrations increased significantly in PLA during the exhaustive exercise, but not in CHO. These two AA are not metabolized in skeletal muscles, and their release from skeletal muscle, measured with arterio-venous difference, is used as a marker of skeletal muscle protein degradation (38). Unfortunately, Blomstrand et al. (39,40) did not report plasma tyrosine and phenylalanine concentrations, but other studies have reported increased plasma concentrations of tyrosine and phenylalanine with exercise. Although increased plasma tyrosine and phenylalanine concentrations during exercise are not established markers of protein degradation, the increased plasma tyrosine and phenylalanine concentration must have resulted from either an increase of their release from tissues or a decrease in their uptake and metabolism in the liver.

Tyrosine and phenylalanine are taken up by the liver at rest, and exercise of an intensity close to that of the present study decreased the absorption of tyrosine and phenylalanine by the liver (41). However, the order of magnitude of the decrease in absorption reported (~1.5 μmol·min^{-1}) cannot fully explain the increase in concentration reported in the present study. In the present study, the increase in concentration of tyrosine and phenylalanine was ~20 μM in PLA, which indicates a total release of ~100 μmol of tyrosine and phenylalanine (with an estimated blood volume of ~5 L). Hence, it seems likely that part of the increase in tyrosine and phenylalanine is the result of increased release from muscles or other tissues.

This release may not come from the skeletal muscles' free amino acid pool, as Essén-Gustavsson and Blomstrand (42) reported that the quantity of tyrosine and phenylalanine increased in this pool during a cycling exercise. In addition, whereas phenylalanine is an essential amino acid and tyrosine is metabolized from phenylalanineand in humans (38), it indicates that a release of these AA from muscles is likely to come from increased protein degradation during exercise.

However, it is important to note that CHO prevented the increase in plasma concentrations of tyrosine and phenylalanine occurring in PLA, which may suggest that carbohydrate intake during exercise reduced protein degradation. Further studies are required to establish the origin of the increase in plasma tyrosine and phenylalanine concentrations during exercise and whether the increase is a valid marker for protein degradation.

Performance. TT performance and gross efficiency were significantly better the day after CHO than after PLA: this indicates the development of less fatigue and/or an improved recovery after the CHO trial. No other parameters (oxygen consumption, RPE, HR, and carbohydrate oxidation) significantly differed between conditions during TT that could explain the difference in performance. Glucose and lactate concentrations were higher after PLA during TT, but the later intake of carbohydrates in PLA compared with CHO during the evening could explain this difference.

Both conditions showed reduced TT performance compared with the rested state measured during familiarization. After CHO, performance was better maintained compared with after PLA. Even if the purpose of this test was familiarization, it still gives some indication of an incomplete recovery on day 2 after both CHO and PLA. Another marker of a poorer recovery after PLA compared with after CHO is the mean power output during the sprint at the start of day 2 (sprint 2): it was also lower in PLA than in CHO, and there was no difference at the start of day 1 (sprint 1). These findings suggest that either carbohydrate intake during the exhaustive exercise on day 1 reduced metabolic stress and the development of fatigue (during or after exercise) or gross efficiency was maintained through maintained muscle function with less protein degradation. To support this, in another study, participants had varying levels of recovery after an exercise to exhaustion after the ingestion of supplements in recovery (10): the results were that the more complete the recovery and the less protein degradation, the better the performance was maintained in a time trial the following day.

Other factors for the maintenance of performance. Significantly higher plasma potassium concentrations on day 2 after PLA compared with after CHO also indicate more pronounced fatigue after PLA than after CHO, as increased plasma potassium concentrations have been identified as a marker of muscle fatigue (43). All this goes toward reduced metabolic stress and development of fatigue.

In addition, Coyle et al. (12) reported no glycogen sparing with carbohydrate intake during an exhaustive exercise. In the present study, participants received the same amount of carbohydrates after the exercise in PLA as during exercise in CHO. The energy intake during day 1 was identical between the two conditions. Therefore, we have no reason to believe glycogen was lower at day 2 in one of the conditions and that this could have explained the difference in TT performance.

Differences in BCAA metabolism. Plasma BCAA concentrations decreased significantly in PLA during the exhaustive exercise, as reported previously (9,10). Interestingly, the plasma BCAA concentrations decreased more in CHO than in PLA. In compliance with our results, previous studies found

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a more pronounced reduction in plasma BCAA concentrations during exercise while consuming carbohydrates compared with a placebo (39,40). However, previous studies reported on smaller differences between the effects of CHO and PLA during exercise than in the present study. This may be because the present study was the first to address the impact during an exhaustive exercise: exhaustive exercise has previously been shown to induce more significant responses in plasma AA concentrations (9,10) than nonexhaustive exercise (39,40,44,45).

One possible explanation for the greater decrease in plasma BCAA concentrations in CHO could be an increased metabolism of BCAA in CHO compared with PLA. BCAA can undergo a transamination with the Krebs cycle intermediate α-ketoglutarate (46). The consumption of carbohydrates during exercise increases the glycolytic flux and, subsequently, the flux of the Krebs cycle (47,48). Consequently, more BCAA can be transaminated with carbohydrate ingestion, which could explain the more significant decrease in plasma BCAA concentrations in CHO.

However, the body has few possibilities to store AA (49), and there was no protein intake before and during the exhaustive exercise. The plasma BCAA must therefore come from endogenous stores of AA: the free AA pool of BCAA in skeletal muscles has been reported to be around 2500–3000 mg (42,45,50). In the present study, the utilization of BCAA represented 77.26 ± 9.11 mg of BCAA in CHO and 42.40 ± 8.76 mg in PLA (condition effect, $P < 0.0001$). Therefore, these quantities could likely have come from the free amino acid pool in muscle and do not imply the need to break down muscle proteins.

Strengths and limitations. The strengths of this study were a single-blinded, balanced crossover design, the inclusion of well-trained cyclists familiarized with the tests, and the standardization of the diet and training the day before and during the 2 d of the protocol. One limitation is the nonrandomization of the order of consumption of drinks. Participants were expected to have a longer time to exhaustion in the CHO condition (12–14,26,27). Hence, the placebo condition had to be performed first to determine the quantity of work they could produce before being exhausted. However, conditions were separated by at least 7 d, and no difference in recovery or performance markers during the incremental fat oxidation profile warm-up or sprint 1 indicates that recovery was incomplete. Additionally, in a similar exercise design, repeating the placebo condition 1 wk after the cycling exercise to exhaustion while ingesting carbohydrate resulted in only a 4% variation in time to exhaustion compared with the first exercise until exhaustion with placebo: this suggests that the order of these exercises only causes a slight variation in the results (12). Another limitation is the absence of muscle biopsies in the design that could enable measurements of muscle glycogen levels during the 2 d of the protocol. Muscle concentrations of free AA could also have been measured to refine estimates of the effects of exercise. Another area for improvement is the relatively low sample number and statistical power of some variables.

CONCLUSIONS

In conclusion, ingesting a carbohydrate drink during an exhaustive exercise reduced deterioration in next-day performance compared with a placebo drink. Maintained gross efficiency explained this performance difference during the time trial. Carbohydrate intake during the exhaustive exercise maintained carbohydrate availability and reduced metabolic stress and the development of fatigue. Carbohydrate intake also induced a better recovery by preventing, to some degree, the muscle damage that occurs during exhaustive exercise. This study highlights the importance of consuming carbohydrates during long-lasting exercise to maintain subsequent performance. Carbohydrate ingestion during exercise is even more critical during competitions over consecutive days.

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All authors have read and approved the final version of the manuscript.

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Data availability statement: The data supporting this study's findings are available from the corresponding author upon reasonable request.

REFERENCES

1. Romijn JA, Coyle EF, Sidossis LS, et al. Regulation of endogenous fat and carbohydrate metabolism in relation to exercise intensity and duration. *Am J Physiol*. 1993;265(3 Pt 1):E380–91.
2. van Loon LJ, Greenhaff PL, Constantin-Teodosiu D, Saris WH, Wagenmakers AJ. The effects of increasing exercise intensity on muscle fuel utilisation in humans. *J Physiol*. 2001;536(Pt 1):295–304.
3. Dahl MA, Areta JL, Jeppesen PB, et al. Coingestion of protein and carbohydrate in the early recovery phase, compared with carbohydrate only, improves endurance performance despite similar glycogen degradation and AMPK phosphorylation. *J Appl Physiol (1985)*. 2020;129(2):297–310.
4. Brooks GA. Amino acid and protein metabolism during exercise and recovery. *Med Sci Sports Exerc*. 1987;19(Suppl 5):S150–6.
5. Rowlands DS, Hopkins WG. Effects of high-fat and high-carbohydrate diets on metabolism and performance in cycling. *Metabolism*. 2002; 51(6):678–90.
6. Lemon PW, Mullin JP. Effect of initial muscle glycogen levels on protein catabolism during exercise. *J Appl Physiol Respir Environ Exerc Physiol*. 1980;48(4):624–9.
7. Décombaz J, Reinhardt P, Anantharaman K, von Glutz G, Poortmans JR. Biochemical changes in a 100 km run: free amino acids, urea, and creatinine. *Eur J Appl Physiol Occup Physiol*. 1979;41(1):61–72.
8. Howarth KR, Phillips SM, MacDonald MJ, Richards D, Moreau NA, Gibala MJ. Effect of glycogen availability on human skeletal muscle protein turnover during exercise and recovery. *J Appl Physiol (1985)*. 2010;109(2):431–8.

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9. Rustad PI, Sailer M, Cumming KT, et al. Intake of protein plus carbohydrate during the first two hours after exhaustive cycling improves performance the following day. *PLoS One*. 2016;11(4):e0153229.
10. Sollie O, Jeppesen PB, Tangen DS, et al. Protein intake in the early recovery period after exhaustive exercise improves performance the following day. *J Appl Physiol (1985)*. 2018;125(6):1731–42.
11. Lemon PW, Orenstein DM, Henke KG. Protein metabolism during moderate exercise in cystic fibrosis patients. *J Sports Sci*. 1984;2(1):47–55.
12. Coyle EF, Coggan AR, Hemmert MK, Ivy JL. Muscle glycogen utilization during prolonged strenuous exercise when fed carbohydrate. *J Appl Physiol (1985)*. 1986;61(1):165–72.
13. Björkman O, Sahlin K, Hagenfeldt L, Wahren J. Influence of glucose and fructose ingestion on the capacity for long-term exercise in well-trained men. *Clin Physiol*. 1984;4(6):483–94.
14. Valentine RJ, Saunders MJ, Todd MK, St Laurent TG. Influence of carbohydrate-protein beverage on cycling endurance and indices of muscle disruption. *Int J Sport Nutr Exerc Metab*. 2008;18(4):363–78.
15. Sjölin J, Stjernström H, Henneberg S, Hambraeus L, Friman G. Evaluation of urinary 3-methylhistidine excretion in infection by measurements of 1-methylhistidine and the creatinine ratios. *Am J Clin Nutr*. 1989;49(1):62–70.
16. Currell K, Jeukendrup AE. Validity, reliability and sensitivity of measures of sporting performance. *Sports Med*. 2008;38(4):297–316.
17. Pritchett K, Bishop P, Pritchett R, Green M, Katica C. Acute effects of chocolate milk and a commercial recovery beverage on postexercise recovery indices and endurance cycling performance. *Appl Physiol Nutr Metab*. 2009;34(6):1017–22.
18. Romano-Ely BC, Todd MK, Saunders MJ, Laurent TS. Effect of an isocaloric carbohydrate-protein-antioxidant drink on cycling performance. *Med Sci Sports Exerc*. 2006;38(9):1608–16.
19. Saunders MJ, Kane MD, Todd MK. Effects of a carbohydrate-protein beverage on cycling endurance and muscle damage. *Med Sci Sports Exerc*. 2004;36(7):1233–8.
20. Péronnet F, Massicotte D. Table of nonprotein respiratory quotient: an update. *Can J Sport Sci*. 1991;16(1):23–9.
21. Kjeldahl J. Neue methode zur bestimmung des stickstoffs in organischen körpern [New method for determining nitrogen in organic bodies]. *Fresenius Zeitschrift F Anal Chemie*. 1883;22(1):366–82.
22. Rowlands DS, Rössler K, Thorp RM, et al. Effect of dietary protein content during recovery from high-intensity cycling on subsequent performance and markers of stress, inflammation, and muscle damage in well-trained men. *Appl Physiol Nutr Metab*. 2008;33(1):39–51.
23. Mariotti F, Tomé D, Mirand PP. Converting nitrogen into protein—beyond 6.25 and Jones' factors. *Crit Rev Food Sci Nutr*. 2008;48(2):177–84.
24. Jeukendrup AE, Wallis GA. Measurement of substrate oxidation during exercise by means of gas exchange measurements. *Int J Sports Med*. 2005;26(Suppl 1):S28–37.
25. Borg GA. Perceived exertion: a note on "history" and methods. *Med Sci Sports*. 1973;5(2):90–3.
26. Coyle EF, Hagberg JM, Hurley BF, Martin WH, Ehsani AA, Holloszy JO. Carbohydrate feeding during prolonged strenuous exercise can delay fatigue. *J Appl Physiol Respir Environ Exerc Physiol*. 1983;55(1 Pt 1):230–5.
27. Kang J, Robertson RJ, Denys BG, et al. Effect of carbohydrate ingestion subsequent to carbohydrate supercompensation on endurance performance. *Int J Sport Nutr*. 1995;5(4):329–43.
28. Børsheim E, Cree MG, Tipton KD, Elliott TA, Aarsland A, Wolfe RR. Effect of carbohydrate intake on net muscle protein synthesis during recovery from resistance exercise. *J Appl Physiol (1985)*. 2004;96(2):674–8.
29. Rand WM, Pellett PL, Young VR. Meta-analysis of nitrogen balance studies for estimating protein requirements in healthy adults. *Am J Clin Nutr*. 2003;77(1):109–27.
30. Young VR, Alexis SD, Baliga BS, Munro HN, Muecke W. Metabolism of administered 3-methylhistidine. Lack of muscle transfer ribonucleic acid charging and quantitative excretion as 3-methylhistidine and its N-acetyl derivative. *J Biol Chem*. 1972;247(11):3592–600.
31. Long CL, Haverberg LN, Young VR, Kinney JM, Munro HN, Geiger JW. Metabolism of 3-methylhistidine in man. *Metabolism*. 1975;24(8):929–35.
32. Williamson DH, Farrell R, Kerr A, Smith R. Muscle-protein catabolism after injury in man, as measured by urinary excretion of 3-methylhistidine. *Clin Sci Mol Med*. 1977;52(5):527–33.
33. Dohm GL, Tapscott EB, Kasperek GJ. Protein degradation during endurance exercise and recovery. *Med Sci Sports Exerc*. 1987;19(5 Suppl):S166–71.
34. Bilmazes C, Uauy R, Haverberg LN, Munro HN, Young VR. Musle protein breakdown rates in humans based on Ntau-methylhistidine (3-methylhistidine) content of mixed proteins in skeletal muscle and urinary output of Ntau-methylhistidine. *Metabolism*. 1978;27(5):525–30.
35. Elia M, Carter A, Bacon S, Winearls CG, Smith R. Clinical usefulness of urinary 3-methylhistidine excretion in indicating muscle protein breakdown. *Br Med J (Clin Res Ed)*. 1981;282(6261):351–4.
36. Rennie MJ, Edwards RH, Krywawych S, et al. Effect of exercise on protein turnover in man. *Clin Sci (Lond)*. 1981;61(5):627–39.
37. Carraro F, Stuart CA, Hartl WH, Rosenblatt J, Wolfe RR. Effect of exercise and recovery on muscle protein synthesis in human subjects. *Am J Physiol*. 1990;259(4 Pt 1):E470–6.
38. Blomstrand E, Saltin B. Effect of muscle glycogen on glucose, lactate and amino acid metabolism during exercise and recovery in human subjects. *J Physiol*. 1999;514(Pt 1):293–302.
39. Blomstrand E, Andersson S, Hassmén P, Ekblom B, Newsholme EA. Effect of branched-chain amino acid and carbohydrate supplementation on the exercise-induced change in plasma and muscle concentration of amino acids in human subjects. *Acta Physiol Scand*. 1995;153(2):87–96.
40. Blomstrand E, Møller K, Secher NH, Nybo L. Effect of carbohydrate ingestion on brain exchange of amino acids during sustained exercise in human subjects. *Acta Physiol Scand*. 2005;185(3):203–9.
41. Felig P, Wahren J. Amino acid metabolism in exercising man. *J Clin Invest*. 1971;50(12):2703–14.
42. Essén-Gustavsson B, Blomstrand E. Effect of exercise on concentrations of free amino acids in pools of type I and type II fibres in human muscle with reduced glycogen stores. *Acta Physiol Scand*. 2002;174(3):275–81.
43. Nielsen JJ, Mohr M, Klarskov C, et al. Effects of high-intensity intermittent training on potassium kinetics and performance in human skeletal muscle. *J Physiol*. 2004;554(Pt 3):857–70.
44. Blomstrand E, Ek S, Newsholme EA. Influence of ingesting a solution of branched-chain amino acids on plasma and muscle concentrations of amino acids during prolonged submaximal exercise. *Nutrition*. 1996;12(7–8):485–90.
45. Galloway SD, Talanian JL, Shoveller AK, Heigenhauser GJ, Spriet LL. Seven days of oral taurine supplementation does not increase muscle taurine content or alter substrate metabolism during prolonged exercise in humans. *J Appl Physiol (1985)*. 2008;105(2):643–51.
46. Wagenmakers AJ, Coakley JH, Edwards RH. Metabolism of branched-chain amino acids and ammonia during exercise: clues from McArdle's disease. *Int J Sports Med*. 1990;11(Suppl 2):S101–13.
47. Gibala MJ, MacLean DA, Graham TE, Saltin B. Tricarboxylic acid cycle intermediate pool size and estimated cycle flux in human muscle during exercise. *Am J Physiol*. 1998;275(2):E235–42.
48. Sahlin K, Jorfeldt L, Henriksson KG, Lewis SF, Haller RG. Tricarboxylic acid cycle intermediates during incremental exercise in healthy subjects and in patients with McArdle's disease. *Clin Sci (Lond)*. 1995;88(6):687–93.
49. Garrow JS. Is there a body protein reserve? *Proc Nutr Soc*. 1982;41(3):373–9.
50. Blomstrand E, Essén-Gustavsson B. Changes in amino acid concentration in plasma and type I and type II fibres during resistance exercise and recovery in human subjects. *Amino Acids*. 2008;37(4):629–36.

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